

DB

DBテーブル一覧

DB制約としてuser_id IS NOT NULL AND guest_session_id IS NULLまたはuser_id IS NULL AND guest_session_id IS NOT NULLのみ許可

項目	内容
repository実装	DB modelとdomain entityを変換する
注意	UsecaseはGORM modelを直接受け取らない

テーブル名	対応Entity	役割
users	User	ログインユーザー、管理者
refresh_tokens	RefreshToken	RefreshToken管理、rotation、reuse検知(もう一度使われたとき)
guest_sessions	GuestSession	未ログインユーザーの行動を一時的に識別
beans	Bean	ランキング対象のコーヒー豆
articles	Article	ランキング対象の記事
bean_articles	BeanArticle	Bean詳細に関連記事を表示するための関連
rank_targets	RankTarget	Bean / Article を共通して扱うための対象
action_events	ActionEvent	閲覧、表示、滞在、クリック、保存、評価、再検索、モーダル操作 CHECK ((user_id IS NOT NULL AND guest_session_id IS NULL) OR (user_id IS NULL AND guest_session_id IS NOT NULL))
modal_display_logs	ModalDisplayLog	推薦モーダルを表示した履歴 CHECK ((user_id IS NOT NULL AND guest_session_id IS NULL) OR (user_id IS NULL AND guest_session_id IS NOT NULL))
modal_block_logs	ModalBlockLog	推薦モーダルを表示しなかった理由 CHECK ((user_id IS NOT NULL AND guest_session_id IS NULL) OR (user_id IS NULL AND guest_session_id IS NOT NULL))
saved_items	SavedItem	UserがBean / Articleをお気に入りとして保存した状態
ratings	Rating	UserがBean / ArticleにGood / Bad評価を付けた状態
content_metrics	ContentMetric	ランキング計算後の集計結果(上記の行動イベントを計算)
interest_profiles	InterestProfile	User / GuestSessionごとの興味スコアとして保存する。どんなものに興味があると思われるか。 CHECK ((user_id IS NOT NULL AND guest_session_id IS NULL) OR (user_id IS NULL AND guest_session_id IS NOT NULL))
batch_runs	BatchRun	ランキング再計算バッチの実行結果
audit_logs	AuditLog	管理者操作、認証上の重要操作、バッチ実行の記録

users

フィールド	Go型	jsonタグ	gormタグ	備考
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"	ユーザーID
Name	string	json:"name"	gorm:"not null"	表示名
Email	string	json:"email"	gorm:"not null;uniqueIndex"	
PasswordHash	string	json:"-"	gorm:"not null"	そのままのパスワード保存禁止。JSONには出さない
Role	UserRole	json:"role"	gorm:"not null;index"	user / admin
Status	UserStatus	json:"status"	gorm:"not null;index"	active / suspended / deleted
TokenVersion	int	json:"-"	gorm:"not null;default:0"	AccessToken強制失効用。外部には基本返さない
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"	作成日時
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null;autoUpdateTime"	更新日時

refresh_tokens

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
UserID	uint64	json:"user_id"	gorm:"not null;index"
TokenHash	string	json:"-"	gorm:"not null;uniqueIndex"
FamilyID	string	json:"-"	gorm:"not null;index"
UsedAt	*time.Time	json:"used_at,omitempty"	gorm:"index"
ReplacedByTokenID	*uint64	json:"replaced_by_token_id,omitempty"	gorm:"index"
RevokedAt	*time.Time	json:"revoked_at,omitempty"	gorm:"index"
ExpiresAt	time.Time	json:"expires_at"	gorm:"not null;index"
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"
User	User	json:"-"	gorm:"foreignKey:UserID;references:ID;constraint"
ReplacedByToken	*RefreshToken	json:"-"	gorm:"foreignKey:ReplacedByTokenID;references:ID;constraint"

guest_sessions

フィールド	Go型	jsonタグ	gormタグ	備考
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"	ゲストセッションID
SessionKeyHash	string	json:"-"	gorm:"not null;uniqueIndex"	もとのguest session keyは保存しない
FirstSeenAt	time.Time	json:"first_seen_at"	gorm:"not null"	初回アクセス日時

LastSeenAt	time.Time	json:"last_seen_at"	gorm:"not null;index"	最終アクセス日時
ExpiresAt	time.Time	json:"expires_at"	gorm:"not null;index"	セッション有効期限

beans

フィールド	Go型	jsonタグ	gormタグ	備考
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"	Bean ID
Name	string	json:"name"	gorm:"not null;index"	豆の表示名
Roaster	*string	json:"roaster,omitempty"	gorm:"index"	焙煎者・焙煎所名
Origin	*string	json:"origin,omitempty"	gorm:"index"	産地
Region	*string	json:"region,omitempty"	-	地域
Farm	*string	json:"farm,omitempty"	-	農園名
Variety	*string	json:"variety,omitempty"	-	品種
RoastLevel	RoastLevel	json:"roast_level"	gorm:"not null;index"	light / medium / dark
Acidity	*int	json:"acidity,omitempty"	-	酸味。1〜5 CHECK (acidity IS NULL OR acidity BETWEEN 1 AND 5)
Bitterness	*int	json:"bitterness,omitempty"	-	苦味。1〜5 CHECK (bitterness IS NULL OR bitterness BETWEEN 1 AND 5)
Flavor	*int	json:"flavor,omitempty"	-	風味。1〜5 CHECK (flavor IS NULL OR flavor BETWEEN 1 AND 5)
Aroma	*int	json:"aroma,omitempty"	-	香り。1〜5 CHECK (aroma IS NULL OR aroma BETWEEN 1 AND 5)
Body	*int	json:"body,omitempty"	-	ボディ。1〜5 CHECK (body IS NULL OR body BETWEEN 1 AND 5)
FlavorNote	*string	json:"flavor_note,omitempty"	-	味や香りなどの説明
Description	*string	json:"description,omitempty"	-	詳細説明。表示時はXSS対策を行う
ImageURL	*string	json:"image_url,omitempty"	-	画像URL
IsPublished	bool	json:"is_published"	gorm:"not null;default:false;index"	公開状態
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"	作成日時
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null;autoUpdateTime"	更新日時

articles

フィールド	Go型	jsonタグ	gormタグ	備考
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"	Article ID
Title	string	json:"title"	gorm:"not null;index"	記事タイトル
Slug	string	json:"slug"	gorm:"not null;uniqueIndex"	URL用識別子
Summary	string	json:"summary"	gorm:"not null"	一覧表示用の短い説明
Body	*string	json:"body,omitempty"	-	記事本文。HTML許可時はsanitize必須
Category	*string	json:"category,omitempty"	gorm:"index"	記事カテゴリ
SourceName	*string	json:"source_name,omitempty"	-	引用元・提供元名

SourceURL	*string	json:"source_url,omitempty"	-	元記事URL
ImageURL	*string	json:"image_url,omitempty"	-	記事画像URL
IsPublished	bool	json:"is_published"	gorm:"not null;default:false;index"	公開状態
PublishedAt	*time.Time	json:"published_at,omitempty"	gorm:"index"	公開日時。下書きならnil
CreatedAt	time.Time	json:"created_at"	gorm:"not null:autoCreateTime"	作成日時
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null:autoUpdateTime"	更新日時

bean_articles

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
BeanID	uint64	json:"bean_id"	gorm:"not null;index;uniqueIndex:uq_bean_articles_bean_article"
ArticleID	uint64	json:"article_id"	gorm:"not null;index;uniqueIndex:uq_bean_articles_bean_article"
DisplayOrder	int	json:"display_order"	gorm:"not null;default:0;index"
CreatedAt	time.Time	json:"created_at"	gorm:"not null:autoCreateTime"
Bean	Bean	json:"-"	gorm:"foreignKey:BeanID;references:ID;constraint:OnUpdate:CASCADE,Or
Article	Article	json:"-"	gorm:"foreignKey:ArticleID;references:ID;constraint:OnUpdate:CASCADE

rank_targets

フィールド	Go型	jsonタグ	gormタグ	備考
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"	RankTarget ID
ContentType	ContentType	json:"content_type"	gorm:"not null;index;uniqueIndex:uq_rank_targets_content"	bean / article
ContentID	uint64	json:"content_id"	gorm:"not null;uniqueIndex:uq_rank_targets_content"	usecaseにて： content_type が bean なら beans テーブルを確認する content_type が article なら articles テーブルを確認する
IsActive	bool	json:"is_active"	gorm:"not null;default:true;index"	ランキング対象として有効か
CreatedAt	time.Time	json:"created_at"	gorm:"not null:autoCreateTime"	作成日時
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null:autoUpdateTime"	更新日時

```
CREATE UNIQUE INDEX uq_rank_targets_content
ON rank_targets (content_type, content_id);
```

action_events

UserID / GuestSessionID の片方だけ許可

状態	保存可否
user_idあり、guest_session_idなし	OK
user_idなし、guest_session_idあり	OK
両方なし	NG
両方あり	NG

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
UserID	*uint64	json:"user_id,omitempty"	gorm:"index"
GuestSessionID	*uint64	json:"guest_session_id,omitempty"	gorm:"index"
EventType	EventType	json:"event_type"	gorm:"not null;index"
RankTargetID	*uint64	json:"rank_target_id,omitempty"	gorm:"index"
Placement	Placement	json:"placement"	gorm:"not null;index"
DwellMs	*int64	json:"dwell_ms,omitempty"	-
RatingScore	*RatingScore	json:"rating_score,omitempty"	-
SearchConditionHash	*string	json:"search_condition_hash,omitempty"	gorm:"index"
PreviousConditionHash	*string	json:"previous_condition_hash,omitempty"	-
SearchKeyword	*string	json:"search_keyword,omitempty"	-
SearchOrigin	*string	json:"search_origin,omitempty"	-
SearchRoastLevel	*RoastLevel	json:"search_roast_level,omitempty"	-
SearchAcidity	*int	json:"search_acidity,omitempty"	-
SearchBitterness	*int	json:"search_bitterness,omitempty"	-
SearchAroma	*int	json:"search_aroma,omitempty"	-
SearchFlavor	*int	json:"search_flavor,omitempty"	-
SearchBody	*int	json:"search_body,omitempty"	-
SearchCategory	*string	json:"search_category,omitempty"	-
ModalDisplayLogID	*uint64	json:"modal_display_log_id,omitempty"	gorm:"index"
PagePath	string	json:"page_path"	gorm:"not null"
ReferrerPath	*string	json:"referrer_path,omitempty"	-

UserAgent	*string	json:"-"	-
IPAddressHash	*string	json:"-"	-
RequestID	*string	json:"request_id,omitempty"	gorm:"index"
OccurredAt	time.Time	json:"occurred_at"	gorm:"not null;index"
User	*User	json:"-"	gorm:"foreignKey:UserID;references:ID;
GuestSession	*GuestSession	json:"-"	gorm:"foreignKey:GuestSessionID;reference NULL"
RankTarget	*RankTarget	json:"-"	gorm:"foreignKey:RankTargetID;reference NULL"
ModalDisplayLog	*ModalDisplayLog	json:"-"	gorm:"foreignKey:ModalDisplayLogID;ref NULL"

modal_display_logs

UserID / GuestSessionID の片方だけ許可

状態	保存可否
user_idあり、guest_session_idなし	OK
user_idなし、guest_session_idあり	OK
両方なし	NG
両方あり	NG

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
UserID	*uint64	json:"user_id,omitempty"	gorm:"index"
GuestSessionID	*uint64	json:"guest_session_id,omitempty"	gorm:"index"
RankTargetID	uint64	json:"rank_target_id"	gorm:"not null;index"
Trigger	ModalTrigger	json:"trigger"	gorm:"not null;index"
PagePath	string	json:"page_path"	gorm:"not null"
ShownAt	time.Time	json:"shown_at"	gorm:"not null;index"
ClickedAt	*time.Time	json:"clicked_at,omitempty"	-
ClosedAt	*time.Time	json:"closed_at,omitempty"	-
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"
User	*User	json:"-"	gorm:"foreignKey:UserID;references:ID;constraint:OnUpda
GuestSession	*GuestSession	json:"-"	gorm:"foreignKey:GuestSessionID;references:ID;constrai NULL"
RankTarget	RankTarget	json:"-"	gorm:"foreignKey:RankTargetID;references:ID;constraint:

```
ALTER TABLE modal_display_logs
ADD CONSTRAINT chk_modal_display_logs_actor
CHECK (
    (user_id IS NOT NULL AND guest_session_id IS NULL)
```

```

OR
(user_id IS NULL AND guest_session_id IS NOT NULL)
);

```

modal_block_logs

UserID / GuestSessionID の片方だけ許可

状態	保存可否
user_idあり、guest_session_idなし	OK
user_idなし、guest_session_idあり	OK
両方なし	NG
両方あり	NG

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	<code>json:"id"</code>	<code>gorm:"primaryKey;autoIncrement"</code>
UserID	*uint64	<code>json:"user_id,omitempty"</code>	<code>gorm:"index"</code>
GuestSessionID	*uint64	<code>json:"guest_session_id,omitempty"</code>	<code>gorm:"index"</code>
CandidateRankTargetID	*uint64	<code>json:"candidate_rank_target_id,omitempty"</code>	<code>gorm:"index"</code>
Reason	ModalBlockReason	<code>json:"reason"</code>	<code>gorm:"not null;index"</code>
PagePath	string	<code>json:"page_path"</code>	<code>gorm:"not null"</code>
BlockedAt	time.Time	<code>json:"blocked_at"</code>	<code>gorm:"not null;index"</code>
User	*User	<code>json:"-"</code>	<code>gorm:"foreignKey:UserID;references:ID"</code>
GuestSession	*GuestSession	<code>json:"-"</code>	<code>gorm:"foreignKey:GuestSessionID;references:ID"</code>
CandidateRankTarget	*RankTarget	<code>json:"-"</code>	<code>gorm:"foreignKey:CandidateRankTargetID;references:ID;onUpdate:CASCADE"</code>

```

ALTER TABLE modal_block_logs
ADD CONSTRAINT chk_modal_block_logs_actor
CHECK (
  (user_id IS NOT NULL AND guest_session_id IS NULL)
  OR
  (user_id IS NULL AND guest_session_id IS NOT NULL)
);

```

saved_items

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	<code>json:"id"</code>	<code>gorm:"primaryKey;autoIncrement"</code>
UserID	uint64	<code>json:"user_id"</code>	<code>gorm:"not null;index;uniqueIndex:uq_saved_items_user_target"</code>
RankTargetID	uint64	<code>json:"rank_target_id"</code>	<code>gorm:"not null;index;uniqueIndex:uq_saved_items_user_target"</code>
RemovedAt	*time.Time	<code>json:"removed_at,omitempty"</code>	<code>gorm:"index"</code>
CreatedAt	time.Time	<code>json:"created_at"</code>	<code>gorm:"not null;autoCreateTime"</code>
UpdatedAt	time.Time	<code>json:"updated_at"</code>	<code>gorm:"not null;autoUpdateTime"</code>
User	User	<code>json:"-"</code>	<code>gorm:"foreignKey:UserID;references:ID;constraint:OnUpdate:CASCADE"</code>

RankTarget	RankTarget	json:"-"	gorm:"foreignKey:RankTargetID;references:ID;constraint:OnUpdate:"

```
CREATE UNIQUE INDEX uq_saved_items_user_target
ON saved_items (user_id, rank_target_id);
```

ratings

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
UserID	uint64	json:"user_id"	gorm:"not null;index;uniqueIndex:uq_ratings_user_target"
RankTargetID	uint64	json:"rank_target_id"	gorm:"not null;index;uniqueIndex:uq_ratings_user_target"
Score	RatingScore	json:"score"	gorm:"not null;index"
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null;autoUpdateTime"
User	User	json:"-"	gorm:"foreignKey:UserID;references:ID;constraint:OnUpdate:CASCADE,C
RankTarget	RankTarget	json:"-"	gorm:"foreignKey:RankTargetID;references:ID;constraint:OnUpdate:CA5

```
ALTER TABLE ratings
ADD CONSTRAINT chk_ratings_score
CHECK (score IN (-1, 1));

CREATE UNIQUE INDEX uq_ratings_user_target
ON ratings (user_id, rank_target_id);
```

```
ALTER TABLE action_events
ADD CONSTRAINT chk_action_events_field_by_type
CHECK (
    (
        event_type IN ('content_view', 'impression', 'click')
        AND rank_target_id IS NOT NULL
        AND dwell_ms IS NULL
        AND rating_score IS NULL
        AND search_condition_hash IS NULL
        AND modal_display_log_id IS NULL
    )
    OR
    (
        event_type = 'stay'
        AND rank_target_id IS NOT NULL
        AND dwell_ms IS NOT NULL
        AND rating_score IS NULL
        AND search_condition_hash IS NULL
        AND modal_display_log_id IS NULL
    )
)
```



```

)
OR
(
    event_type = 'rating'
    AND rank_target_id IS NOT NULL
    AND rating_score IN (-1, 1)
    AND dwell_ms IS NULL
    AND search_condition_hash IS NULL
    AND modal_display_log_id IS NULL
)
OR
(
    event_type = 'save'
    AND rank_target_id IS NOT NULL
    AND dwell_ms IS NULL
    AND rating_score IS NULL
    AND search_condition_hash IS NULL
    AND modal_display_log_id IS NULL
)
OR
(
    event_type = 're_search'
    AND rank_target_id IS NULL
    AND dwell_ms IS NULL
    AND rating_score IS NULL
    AND search_condition_hash IS NOT NULL
    AND modal_display_log_id IS NULL
)
OR
(
    event_type IN ('modal_impression', 'modal_click', 'modal_close')
    AND rank_target_id IS NOT NULL
    AND dwell_ms IS NULL
    AND rating_score IS NULL
    AND search_condition_hash IS NULL
    AND modal_display_log_id IS NOT NULL
)
);

```

content_metrics

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
RankTargetID	uint64	json:"rank_target_id"	gorm:"not null;uniqueIndex"
Score	float64	json:"score"	gorm:"not null;default:0;index"
ImpressionCount	int64	json:"impression_count"	gorm:"not null;default:0"
ContentViewCount	int64	json:"content_view_count"	gorm:"not null;default:0"
ClickCount	int64	json:"click_count"	gorm:"not null;default:0"
StayTotalMs	int64	json:"stay_total_ms"	gorm:"not null;default:0"
SaveCount	int64	json:"save_count"	gorm:"not null;default:0"
RatingCount	int64	json:"rating_count"	gorm:"not null;default:0"
GoodCount	int64	json:"good_count"	gorm:"not null;default:0"
BadCount	int64	json:"bad_count"	gorm:"not null;default:0"
RatingAvg	float64	json:"rating_avg"	gorm:"not null;default:0"

フィールド	Go型	jsonタグ	gormタグ
GoodRate	float64	json:"good_rate"	gorm:"not null;default:0"
BadRate	float64	json:"bad_rate"	gorm:"not null;default:0"
ModalImpressionCount	int64	json:"modal_impression_count"	gorm:"not null;default:0"
ModalClickCount	int64	json:"modal_click_count"	gorm:"not null;default:0"
ModalCloseCount	int64	json:"modal_close_count"	gorm:"not null;default:0"
ClickRate	float64	json:"click_rate"	gorm:"not null;default:0"
SaveRate	float64	json:"save_rate"	gorm:"not null;default:0"
ModalClickRate	float64	json:"modal_click_rate"	gorm:"not null;default:0"
ModalCloseRate	float64	json:"modal_close_rate"	gorm:"not null;default:0"
PeriodStart	time.Time	json:"period_start"	gorm:"not null;index"
PeriodEnd	time.Time	json:"period_end"	gorm:"not null;index"
CalculatedAt	time.Time	json:"calculated_at"	gorm:"not null;index"
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null;autoUpdateTime"
RankTarget	RankTarget	json:"-"	gorm:"foreignKey:RankTargetID;references:ID;constrai

```
CREATE UNIQUE INDEX uq_content_metrics_rank_target
ON content_metrics (rank_target_id);
```

re_search_countは持たない。
re_searchはRankTarget に直接紐づかないため、content_metrics で集計しない。

interest_profiles

UserID / GuestSessionID の片方だけ許可

状態	保存可否
user_idあり、guest_session_idなし	OK
user_idなし、guest_session_idあり	OK
両方なし	NG
両方あり	NG

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
UserID	*uint64	json:"user_id,omitempty"	gorm:"index"
GuestSessionID	*uint64	json:"guest_session_id,omitempty"	gorm:"index"
Dimension	InterestDimension	json:"dimension"	gorm:"not null;index"
Value	string	json:"value"	gorm:"not null;index"
Score	float64	json:"score"	gorm:"not null;default:0;index"
LastEventAt	time.Time	json:"last_event_at"	gorm:"not null;index"
ExpiresAt	*time.Time	json:"expires_at,omitempty"	gorm:"index"
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime"
UpdatedAt	time.Time	json:"updated_at"	gorm:"not null;autoUpdateTime"
User	*User	json:"-"	gorm:"foreignKey:UserID;references:ID;constraint:OnUp

GuestSession	*GuestSession	json:"-"	gorm:"foreignKey:GuestSessionID;references:ID;constr
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```

ALTER TABLE interest_profiles
ADD CONSTRAINT chk_interest_profiles_actor
CHECK (
    (user_id IS NOT NULL AND guest_session_id IS NULL)
    OR
    (user_id IS NULL AND guest_session_id IS NOT NULL)
);

CREATE UNIQUE INDEX uq_interest_profiles_user_dimension_value
ON interest_profiles (user_id, dimension, value)
WHERE user_id IS NOT NULL;

CREATE UNIQUE INDEX uq_interest_profiles_guest_dimension_value
ON interest_profiles (guest_session_id, dimension, value)
WHERE guest_session_id IS NOT NULL;

```

batch_runs

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
JobName	string	json:"job_name"	gorm:"not null;index"
Status	BatchStatus	json:"status"	gorm:"not null;index"
StartedAt	time.Time	json:"started_at"	gorm:"not null;index"
FinishedAt	*time.Time	json:"finished_at,omitempty"	-
RowsProcessed	int64	json:"rows_processed"	gorm:"not null;default:0"
ErrorMessage	*string	json:"error_message,omitempty"	-
TriggeredBy	AuditActorType	json:"triggered_by"	gorm:"not null;index"
TriggeredUserID	*uint64	json:"triggered_user_id,omitempty"	gorm:"index"
TriggeredUser	*User	json:"-"	gorm:"foreignKey:TriggeredUserID;references:ID;constraint:ON UPDATE CASCADE,ON DELETE CASCADE"

audit_logs

フィールド	Go型	jsonタグ	gormタグ
ID	uint64	json:"id"	gorm:"primaryKey;autoIncrement"
ActorType	AuditActorType	json:"actor_type"	gorm:"not null;index"
ActorUserID	*uint64	json:"actor_user_id,omitempty"	gorm:"index"
Action	AuditAction	json:"action"	gorm:"not null;index"
TargetType	*string	json:"target_type,omitempty"	gorm:"index"
TargetID	*uint64	json:"target_id,omitempty"	gorm:"index"
Detail	*string	json:"detail,omitempty"	-

IPAddressHash	*string	json:"-"	-
RequestID	*string	json:"request_id,omitempty"	gorm:"index"
CreatedAt	time.Time	json:"created_at"	gorm:"not null;autoCreateTime;index"
ActorUser	*User	json:"-"	gorm:"foreignKey:ActorUserID;references:ID;constraint:OnU NULL"

DB制約・Index 実装ルール

以下はGORMタグだけに依存せず、migration SQLで明示する。

action_events

```
ALTER TABLE action_events
ADD CONSTRAINT chk_action_events_actor
CHECK (
    (user_id IS NOT NULL AND guest_session_id IS NULL)
    OR
    (user_id IS NULL AND guest_session_id IS NOT NULL)
);
```

```
ALTER TABLE action_events
ADD CONSTRAINT chk_action_events_dwell_ms
CHECK (
    dwell_ms IS NULL
    OR
    (dwell_ms >= 3000 AND dwell_ms <= 1800000)
);
```

```
ALTER TABLE action_events
ADD CONSTRAINT chk_action_events_rating_score
CHECK (
    rating_score IS NULL
    OR
    rating_score IN (-1, 1)
);
```